

Body/Equipment Mounting Directives for Vans



MBAC

Mercedes-Benz Advanced Control

Additional Directive for Mercedes-Benz Advanced Control (MBAC)

Edition AeJ2021-1a

Supplement to the Body/Equipment Mounting Directives for
Sprinter BR 907 and BR 910

Editions AeJ2021-1a

Mercedes-Benz



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1.1 Aim of this Directive

The present additional directive for the Mercedes-Benz Advanced Control (MBAC) is directed at professional body manufacturers who use the MBAC for their respective body solutions. As a result, this additional directive assumes that the body manufacturer has suitable background knowledge.

The document may only be used as a complete document and it is valid as a supplement to the vehicle-specific body/equipment mounting directives. Excerpts are not permitted.

The issues and information outlined here apply only in combination with one of the following documents:

- Body/equipment mounting directive BR 907, current edition AeJ2021-1a
- Body/equipment mounting directive BR 910, current edition AeJ2021-1a

Consult these vehicle-specific body/equipment mounting directives for any issues not outlined here.

The additional directive for the MBAC and the vehicle-specific body/equipment mounting directives, as well as other technical information and documents, can be found in the Bodybuilder Portal at the following Internet address:

<https://bb-portal.mercedes-benz-vans.com>

In addition, consult the valid operator's manual for the respective vehicle and ensure that this is adhered to.

You can download operator's manuals online from the relevant country-specific Internet site of Mercedes-Benz AG.

If necessary, get in touch with the contact person at Mercedes-Benz Service.

Information and specifications to be adhered to regarding vehicle modifications by the body manufacturer and the subsequent vehicle clearing by recognized appraisers or inspectors can be found in the vehicle-specific body/equipment mounting directive.

Inspections conducted by official test centers or official approvals do not inevitably guarantee compatibility with all the functions and systems in the basic vehicle.

Should you have any queries about compatibility validation or if you require information about it, please contact the technical advice service for body manufacturers.

1.2 Conventions

The layout and conventions used in this document are the same as those of the body/equipment mounting directives.

1.3 Overview of MBAC

1.3.1 What is the MBAC?

MBAC stands for "Mercedes-Benz Advanced Control". The MBAC is a control unit and interface module that removes the separation between the base vehicle and the living area. It networks the components before and after the B-pillar while also enabling easy and comfortable operation. Thanks to the MBAC, camper vans based on Mercedes-Benz vans can become a connected home on wheels.

The customer intuitively controls the MBAC either through the MBUX multimedia system, an optional 10.25-inch display that can be positioned anywhere in the body, or with their smartphone.

The MBAC supports all conventional smartphones.

1.3.2 Variants

The integrated control concept using smartphone, additional display and head unit (Mercedes-Benz User Experience MBUX) is available to you in different variants. Along with the MBAC, code JA3, the parameterizable special module (PSM)/multi-purpose module (MPM), code ED5, is also absolutely essential. All other equipment can be selected as an option.

MBAC variants with equipment

MBAC Entry (level 1)	MBAC High (level 2)	MBAC Max (level 3)
Control via smartphone only	Control via smartphone and MBUX • MBUX (code E4M)	Control via smartphone, additional display and MBUX • Additional display (code JH7) via Global Logistics Center • MBUX (code E4M)

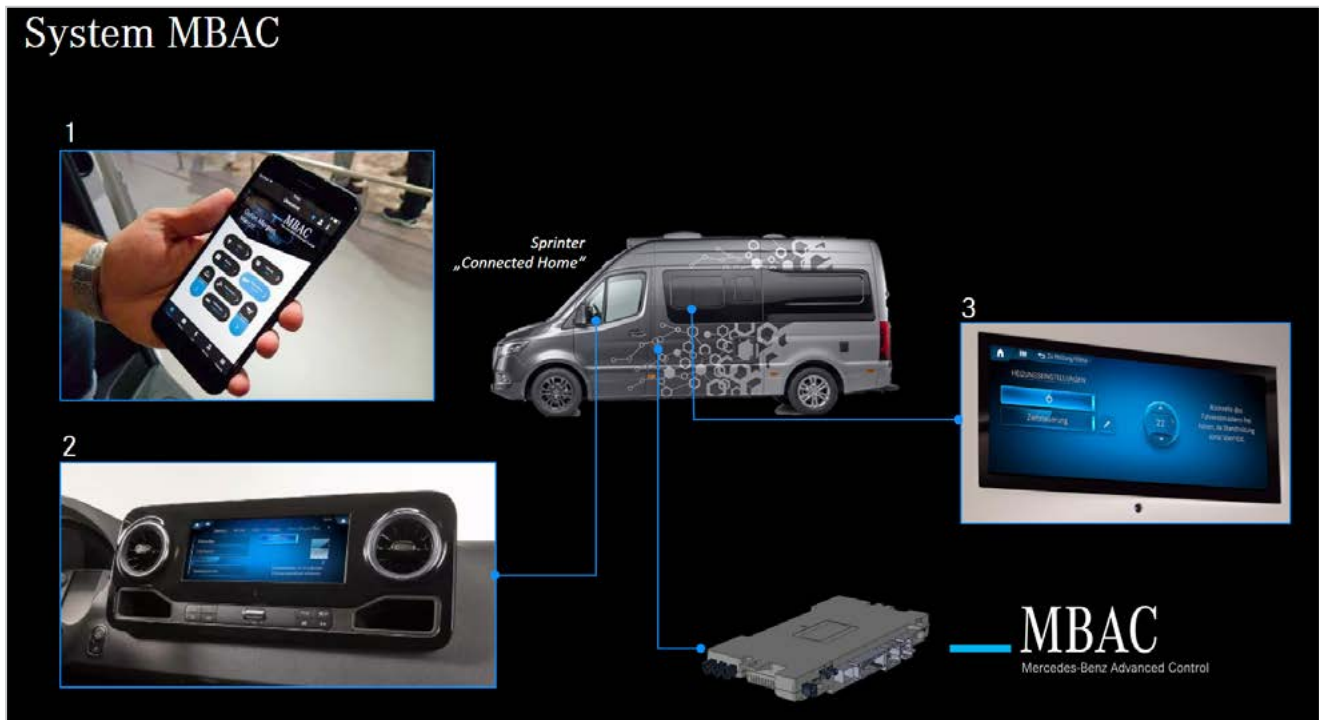
Ordering options with codes

MBAC	JA3	MBAC	JA3	MBAC	JA3
PSM (MPM) (obligatory)	ED5	PSM (MPM) (obligatory)	ED5	PSM (MPM) (obligatory)	ED5
Auxiliary battery cutoff relay	E36	Auxiliary battery cutoff relay	E36	Auxiliary battery cutoff relay	E36
Optional		–		Additional display pre-installation	JH7
		MBUX	E4M	MBUX	E4M

MBAC	Mercedes-Benz Advanced Control
PSM	Parameterizable special module
MPM	Multi-Purpose Module (Mercedes-Benz-internal designation for PSM)
MBUX	Head unit with Mercedes-Benz User Experience

1.3.3 Basic Concept and Requirements

Depending on the selected MBAC and base vehicle variant, the vehicle corresponds to the following schematic.



The MBAC is the Mercedes-Benz smarthome controller unit for camper vans. It can be operated via the "Mercedes-Benz Advanced Control" smartphone app (see chapter 5 MBAC app (→ page 36)), via MBUX and via the additional display in the living area.

- 1 MBAC control via smartphone app
- 2 MBAC operation through MBUX possible
- 3 The additional display can be installed in the living area by the body manufacturer.

The following work must be conducted first, before the start of MBAC production at the body manufacturer and before the first vehicle is ordered:

1. MBAC concept integration at body manufacturer

- All relevant MBAC scopes must be integrated on the concept side at the body manufacturer; this concerns:
 - MBAC
 - Bluetooth® pushbutton switch
 - Extension module
 - Additional display

For this, see Chapter 2 Installation Specifications (→ page 10).

2. Component check

- The terminals controlled by the MBAC and installed by the body manufacturer are to be validated through a compatibility test with Mercedes-Benz MBAC Support in the Mercedes-Benz Vans Body Manufacturer Center.

3. Commissioning the MBAC

- The MBAC can be commissioned through a separate XENTRY Kit or at a Mercedes-Benz workshop/company-owned sales and service outlet (see Chapter 6.1 Commissioning (→ page 37)).
- In accordance with the respective body configuration(s), a specific configuration file is required that is transferred to the vehicle per XENTRY Kit. The configuration file(s) is/are to be coordinated with MBAC Support in the Mercedes-Benz Vans Body Manufacturer Center and subsequently made available.

For completion, perform the following operations:

- ▶ The MBAC is supplied ex factory installed in the seat base. The MBAC must be installed at a new position as per the specifications at Chapter 2.1.2 Material, installation position and protection class (→ page 10).
- ▶ Depending on where the new installation position for the MBAC is located, it may be necessary to extend the wiring harness between the MBAC and the PSM (MPM).
- ▶ Optional: Connection of the additional display, preparation of a separate wiring harness.
- ▶ Optional: Installation of the additional display and an extra pushbutton switch.
- ▶ Connection of separate components as required.
- ▶ Configuration of MBAC using XENTRY Kit.
- ▶ Commissioning.

1.4 Product safety and product liability

1.4.1 Product safety

Vehicle manufacturers and body manufacturers alike must always ensure that the products they manufacture are safe when they are brought into circulation and do not present any danger to persons or property. Otherwise there may be consequences under civil, criminal or administrative law. Every manufacturer is always liable for the product that it has manufactured.

The body manufacturer must ensure compliance with Directive 2001/95/EC concerning general product safety.

In addition, the body manufacturer must ensure adherence to functional reliability standards (as per ISO 26262, see Chapter 1.4.3 Functional reliability (→ page 7)) and cyber security standards (as per ISO 21434 and UN-R155).

1.4.2 Product liability

The body manufacturer bears responsibility for the operational and road safety of the body mounting work performed by him, and in particular for:

- testing and maintaining the operating and driving safety of the overall vehicle after the body is mounted (the body must not have a negative effect on the driving, braking or steering characteristics of the vehicle)
- the effects of body mounting work on the chassis
- consequential damage arising from body mounting work
- consequential damage resulting from retrofitted electrical and electronic systems
- maintaining the functional reliability and unobstructed movement of all moving parts of the basic vehicle (e.g. axles, springs, propeller shafts, steering, gearshift linkage, etc.) after the body mounting work is complete; even in the case of diagonal torsion of the vehicle

Work carried out or modifications to the basic vehicle or body must be entered in the maintenance booklet in the "Confirmations of the body manufacturer" section.

1.4.3 Functional reliability

With the MBAC, an interface has been made available to the body manufacturer, not a system. The MBAC functionalities made available through the interface are safeguarded as per ISO 26262 in accordance with level QM. The body manufacturer is responsible at all times for safeguarding the functions in which the MBAC is involved, and ensuring that established standards (e.g. ISO 26262 on functional reliability) and specifications are complied with.

1.4.4 Assessment of Known Functions

Functions that Mercedes-Benz is already familiar with have been assessed with regard to their safety relevance. This does not discharge the manufacturer of the end product from its liability and obligation to conduct its own tests and analyses on functional reliability in accordance with good engineering practice.

Status information

To display the following status information in installed displays, read-only access is granted to the vehicle status.

The following indicators are taken into consideration:

- Doors and flaps
- Water tank
- Battery management
- Gas level
- Outside temperature sensor
- Solar module

Comfort functions

Mercedes-Benz considers the following functions to be purely comfort functions.

The following operation elements are taken into consideration:

- Air conditioning system
- Lighting circuit/ambiance
- Motion detector
- TV drawer
- Sound system
- Water pump

Actuator functions

The following functions are actuator functions with potential safety relevance. A final assessment of the safety relevance however, can only be provided by the manufacturer of the overall system.

The following operation elements are taken into consideration:

- Windows and roof hatches
- Access step
- Sat operation

- Roller blinds

A speed check is required for this function for the windshield and side windows.

Rear roller sunblinds that do not impair the driver's view can be excepted here.

- Awning
- Water drain
- Slide-out (extendable body window)

To safeguard these functions Mercedes-Benz enables the actuator functions to be locked when driving. To do so, the actuator scopes must be locked at a speed >0 km/h in the MBAC using the Config file. This safeguarding is in accordance with level QM as per ISO 26262.

If a higher safeguarding level is required, then this must be implemented by the body manufacturer. Safeguarded signals (speed/ignition) can be used by the PSM (MPM) for this; for more information, see the PSM/MPM brochure in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>

1.5 Contact person

Technical advice service for the MBAC

- ① For technical support queries, please contact the technical advice service for the MBAC (in German or English) via the following email address:

technical_support_mbac@daimler.com

- ① Further information on persons to contact for advice concerning the basic vehicle and its equipment, as well as the PSM (MPM), can be found in the vehicle-specific body/equipment mounting directives.

2.1 Base module

The MBAC base module is installed at the factory, provisionally in the seat base. The housing of the MBAC serves to protect the printed circuit board and it is two-piece (upper and lower part). For cooling of the MBAC, the upper part of the housing contains louvers or a cutout for a heat sink (passive cooling) that is seated on the printed circuit board.

2.1.1 Weight and dimensions

The weight of the MBAC is ≤ 700 g.

The maximum dimensions of the housing are:

$L \times B \times H = 280 \times 160 \times 40$ mm

2.1.2 Material, installation position and protection class

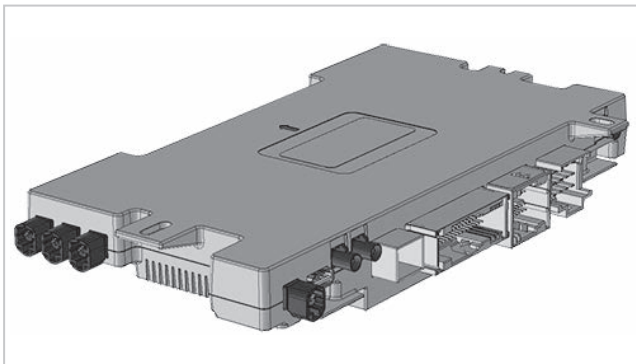
The housing of the MBAC is made of polypropylene and this material is classified in accordance with the fire protection procedure UL94 (Fire Protection).

The MBAC can be installed horizontally or vertically in a body-manufacturer-specific vehicle body or conversion.

Horizontal installation

A horizontally installed housing has the protection class IP31.

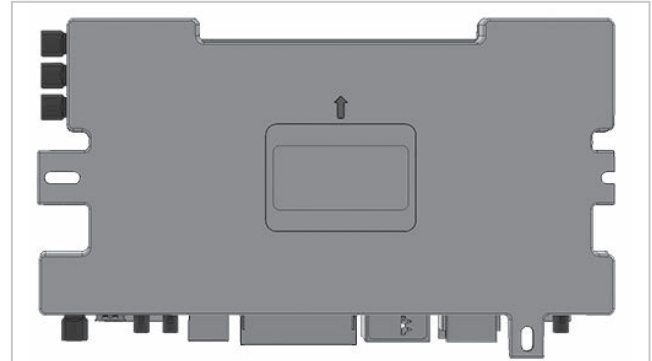
To guarantee protection class IP31 for a horizontal installation, the MBAC must be installed with the closed side of the housing facing upwards.



Horizontal installation position

Vertical installation

A vertically installed housing has the protection class IP30. It must be ensured here that the arrow on the housing faces upwards when in installed position.



Vertical installation position

2.1.3 Operating temperature

The minimum ($T_{\min}$) and maximum ($T_{\max}$) operating temperatures of the MBAC and of the components with overload/low-temperature protection are:

$T_{\min} = -30$ °C

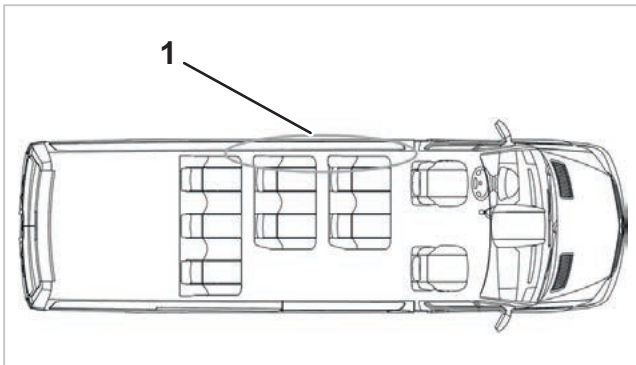
$T_{\max} = +80$ °C

2.1.4 Wireless Interface

The MBAC is equipped with a Bluetooth® wireless interface (Standard 4.2) for the connection of an external antenna. This connection is provided through the plug connector model Rosenberger FAKRA 59S2UQ-40MT5-K. The coding of the Bluetooth® antenna connection is Curry (K).

Antenna installation position

The antenna must be positioned so that the best-possible reception is ensured. The labeled side of the antenna may not be covered up by metalized parts. The labeled side of the antenna is the emission side and it should be aligned, where possible, towards the living area door. Ideally, on the opposite side of the vehicle between the B-pillar and C-pillar.



Antenna installation position

1 Area for ideal position of antenna

2.1.5 Service Interface

The MBAC provides a CAN interface for the XENTRY service.

2.1.6 Service Accessibility

- ▶ The MBAC is to be installed so that it can be easily accessed in a service case.
- ▶ The installation position is to be documented in the operator's manual by the body manufacturer.

The installation of the MBAC control unit must not impair the prescribed times allowed by Mercedes-Benz for maintenance operations.

2.1.7 Electrical connection to the basic vehicle

! NOTE

The MBAC must be connected to a body-manufacturer-specific auxiliary battery.

Only electrically connect the auxiliary battery with the main battery of the basic vehicle via an appropriately designed battery cutoff relay. The body manufacturer is responsible for ensuring a positive charge balance for all installed components in this way, as well as a maximum permissible total charging current of 80 A.

- ① Further information on this and the specifications to be adhered to can be found in the vehicle-specific body/equipment mounting directive.

2.2 Extension module

2.2.1 Function and Connection

If additional interfaces are required other than those described under sections 3 and 4, the MBAC extension module is then used. This is identical to the MBAC base module (see chapter 2.1 Base module (→ page 10)).

When retrofitting the MBAC extension module, the MBAC vehicle plug must be disconnected so that the system is de-energized.

The power supply to the coding pins means that, when connecting the MBAC, care must be taken to ensure that the body 1 plug (see chapter 3.2.1 MBAC slots on longer side (→ page 20)) is connected before the body battery plug. The extension module can be installed irrespective of where the MBAC is installed. The installation is identical to that of the MBAC base module, see chapter 2.1.2 Material, installation position and protection class (→ page 10). The wiring harness manufacturer must ensure here that CAN communication runs smoothly.

The MBAC is variant-coded through the wiring harness:

Coding input bit 0 > (Coding 0)	Coding input bit 1 (Coding 1)	Coding	IO Board Diagnosis Postfix
0	0	0	_EXT1
1	0	1	_EXT2
0	1	2	_EXT3
1	1	0	_EXT1

Function coding for the base module (EXT1) and extension module (EXT2 or EXT3) is conducted by assigning the Coding 0 and Coding 1 (body plug 1 pin 45/46) inputs to the Ground (Gnd) level or the supply voltage in accordance with the table.

Communication between the modules is then over the MBAC-CAN. The CAN termination is configured on the base module using DIP switches.

The extension module is set to "open". The base module is set according to whether an additional display is present.

- Without additional display: 60 ohms
- With additional display, 120 ohms

The extension module can be installed irrespective of where the MBAC is installed.

2.3 Additional display

The additional display can be positioned anywhere in the body. The ventilation slots on the rear panel should not be covered up in doing so. Cooling is not required here because the display is equipped with an integrated temperature sensor and at high temperatures, the background illumination of the display is regulated.

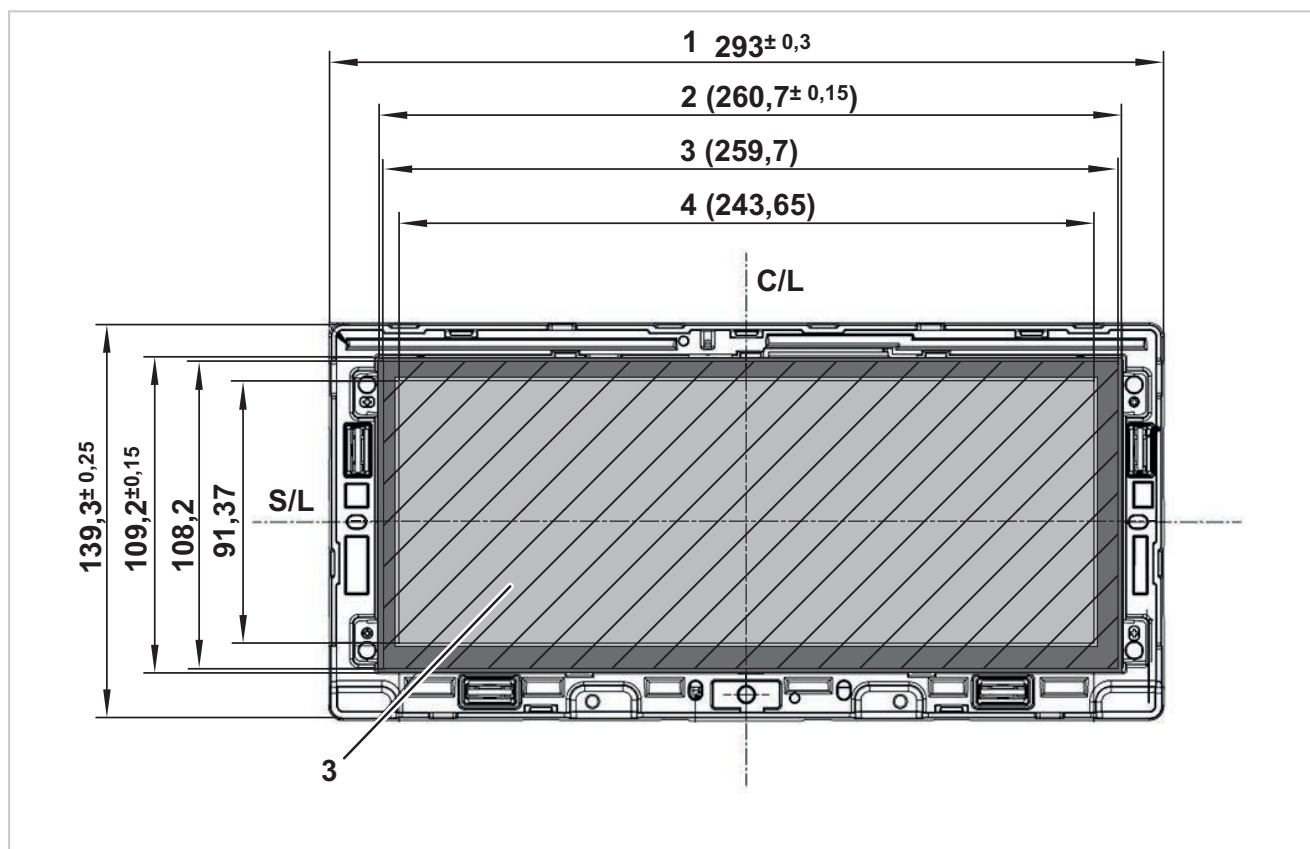


2.3.1 Specifications

- Weight: 1.1 kg
- Resolution: 1920 pix (H) × 720 pix (V) × 3 (RGB)
- Active area: 243.65 × 61.37 mm
- Display size: 10.25"
- Color depth: 8 bit, 16777216 colors
- Cover glass: Corning Gorilla 2320, 0.7±0.05 mm

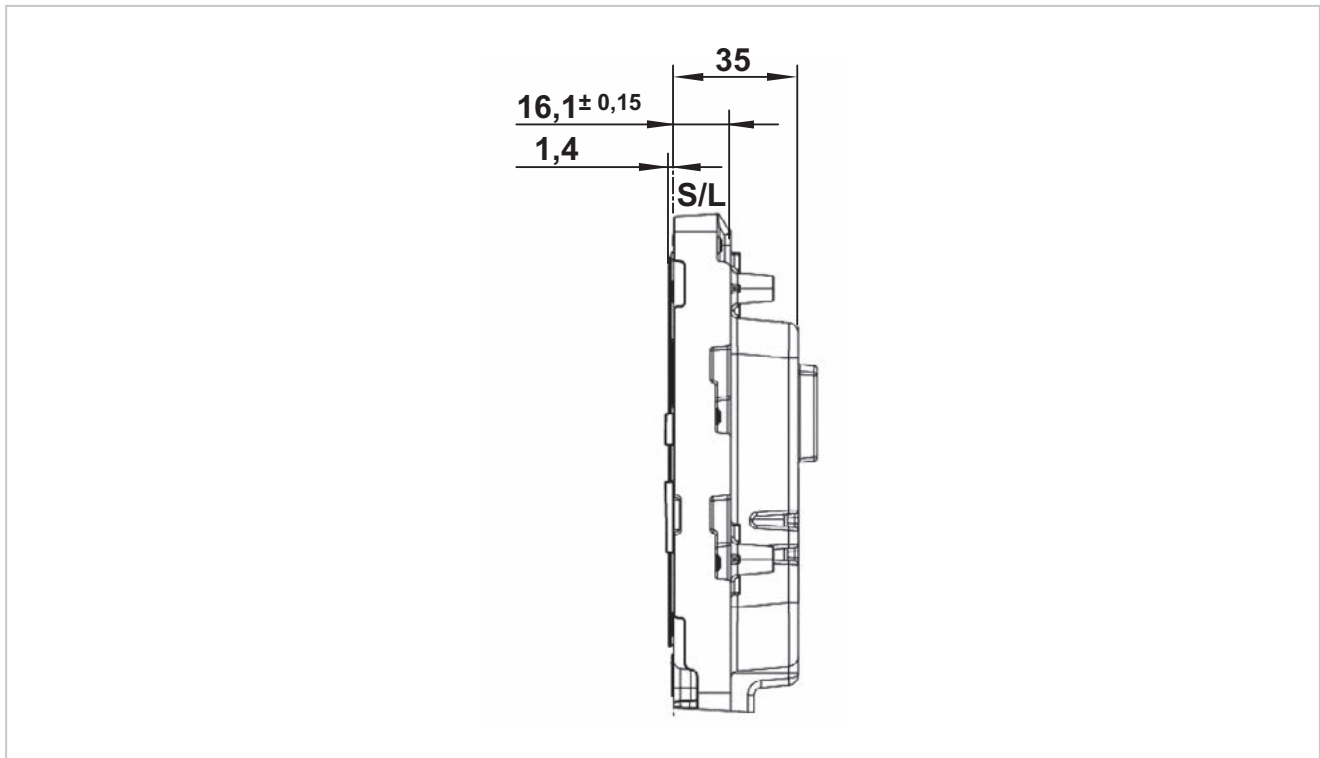
2.3.2 Drawing, dimensional concept

Dimensional pictures for additional display



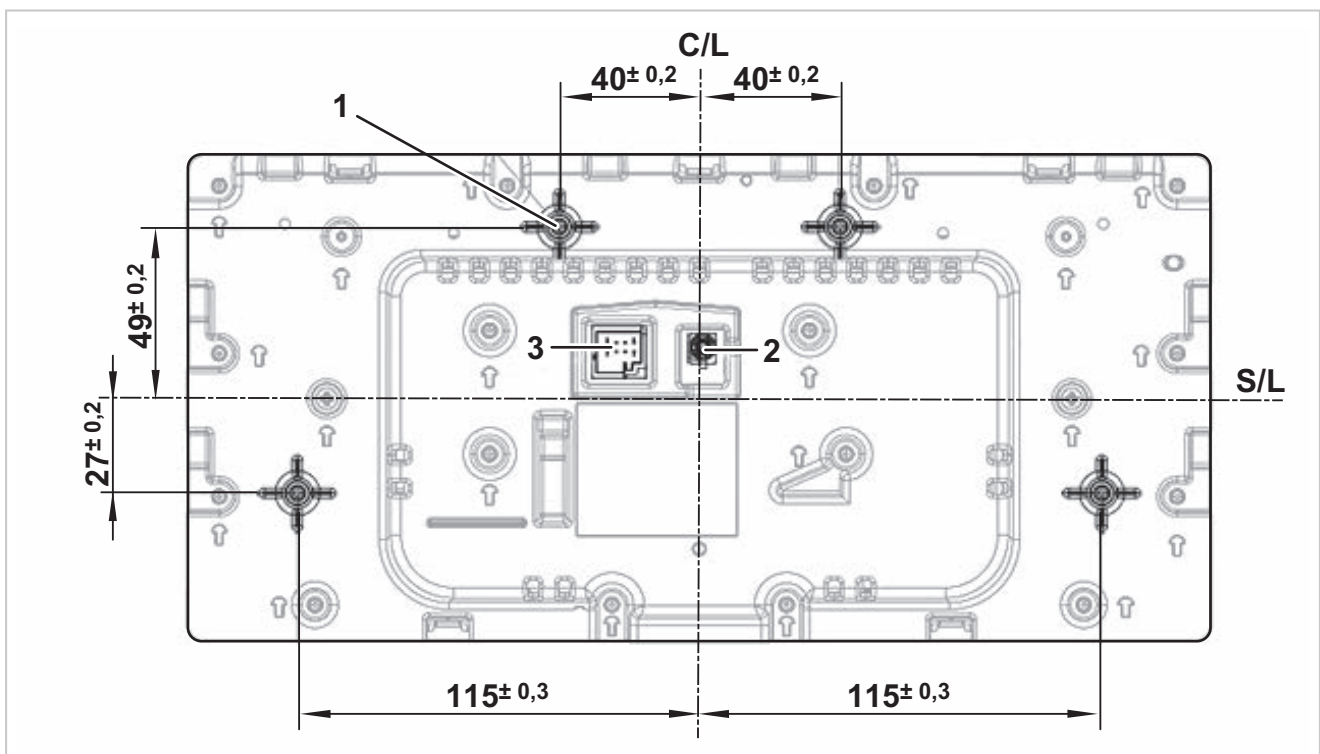
Additional display front view, dimensions in mm

- 1 Housing exterior dimension
- 2 Safety glass
- 3 Hatched: Protective film, material ST-6500, transparent (colorless)
- 4 Active picture area (light gray)



Additional display side view, dimensions in mm

Optional screw/fastening points



Additional display rear view

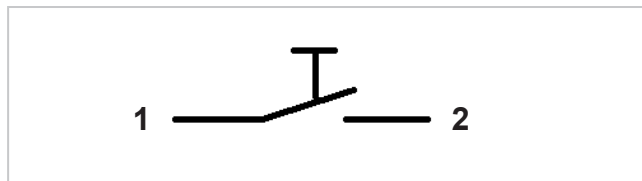
- 1 Fastening points for 4 screws M5×12 mm
- 2 Connection for video signal through FAKRA connector
- 3 Connection for power supply and CAN through CID Interface Connector

2 Installation Specifications

2.3.3 Additional Wake-Up Switches

The additional display is activated using a pushbutton switch.

The pushbutton switch can be selected from the body manufacturer's switch range.



Normally-open contact to ground on vehicle plug

- 1 Pin 9
- 2 Pin 8 (ground)

2.3.4 Connections, cables, E/E

The additional display communicates with the base module over the MBAC CAN.

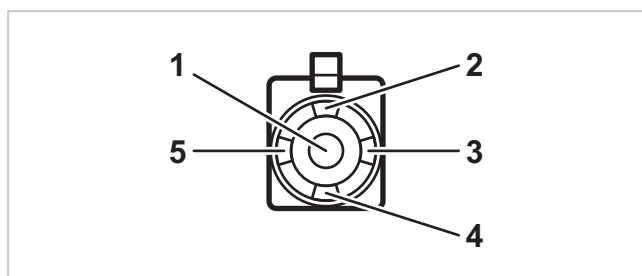
The additional display has an integrated CAN termination. Therefore the CAN termination must be configured on the base module using DIP switches.

- Without additional display: 60 ohms
- With additional display: 120 ohms

The lines must be kept as short as possible. The length of the video line is particularly critical. For longer distances, a high-quality line with excellent conductivity properties and shielding with good attenuation is required.

The additional display is equipped with two connections:

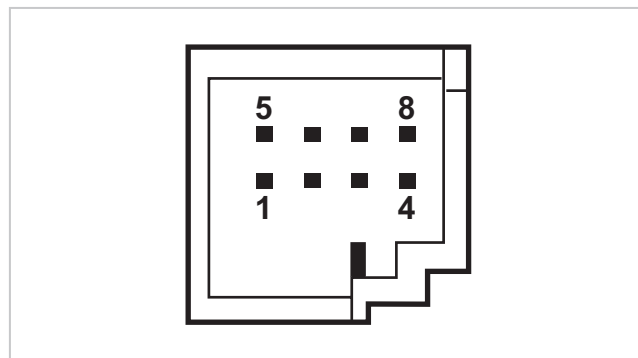
Connection for video signal input from the MBAC



FAKRA Connector MOLEX

	Assignment
1	HSV L+
2	GND
3	GND
4	GND
5	GND

Connection for power supply and CAN



CID Interface Connector

	Assignment	
1	Terminal_31 (GND)	5 UART_Rx
2	Terminal_30 (Power)	6 NC
3	NC	7 UART_Tx
4	CAN low	8 CAN high

Lines

A perfect function is guaranteed with a line length of up to 10 m for the video signal (connection to FAKRA connector).

Bluetooth® antenna

A Bluetooth® antenna supports communication with compatible smart devices. The Bluetooth® antenna can be connected to the Bluetooth® interface of the MBAC and be used for testing.

The Bluetooth® antenna has the following properties:

- Return loss of the antenna connection: ≤ -10 dB
- Load capacity: 1 W
- Voltage Standing Wave Ratio (VSWR): max 2:1
- Impedance: 50 ohms
- Minimum/maximum gain in installed status: 0.0 dBi/+3.0 dBi
- Frequency range: 2400–2500 MHz

3.1 Port Assignment

3.1.1 Performance Values

The following power is required by the MBAC depending on the given operating status:

- Ignition off/video mode off: Medium power requirement of 1.5 mA
- Video mode standby: 200 mA
- Video mode active: 1000 mA

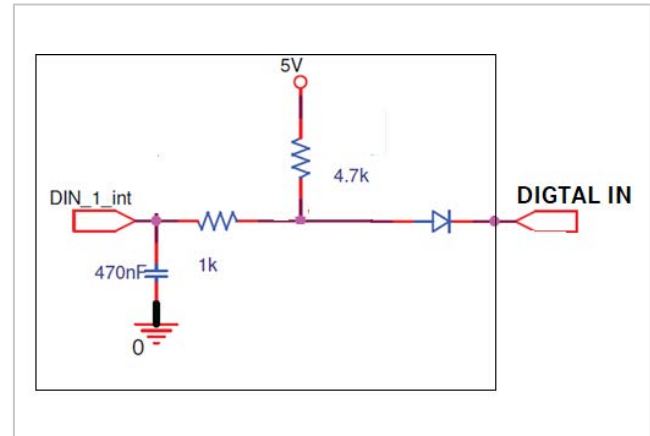
The maximum total switching capacity for the MBAC is 350 W.

- i** Information on the electrical connection to the basic vehicle can be found in Chapter 2.1.7 Electrical connection to the basic vehicle (→ page 11).

3.1.2 Digital inputs and outputs, PWM outputs

Digital inputs

The digital inputs are used to read in digital signals (0 XOR 1), and they can be read out by the embedded software.



ON Input voltage $V_{IN} < 1.8 \text{ V}$

OFF Input voltage $V_{IN} > 2.8 \text{ V}$

The MBAC can operate up to 29 digital inputs. Of these, there are 16 specific digital inputs with wake-up capability and 13 specific digital inputs with no wake-up capability. Along with the 29 digital inputs, there is another wake-up-capable input for reading in the Bluetooth® pairing pushbutton switch. All digital inputs are operated as "low active" ones, i.e. connected pushbutton switches switch to ground. The LED in the Bluetooth® pairing pushbutton switch is powered by the low-power PWM output and it can be activated with a maximum of 10 mA.

Digital outputs

The digital outputs are used to set digital signals (0 XOR 1), and they can be set by the embedded software.

The functionality of the digital outputs is provided by the driver modules TLE94110EL. The MBAC can operate up to 29 digital outputs. These can provide a rated current of 200 mA and 12 V (High Side). Along with the 29 digital outputs, there is another output for the MPM wake-up signal that has a maximum output resistance of 1 kOhm, can supply a maximum output current of 30 mA and can switch 12 V (High Side). The MBAC can operate 14 PWM outputs.

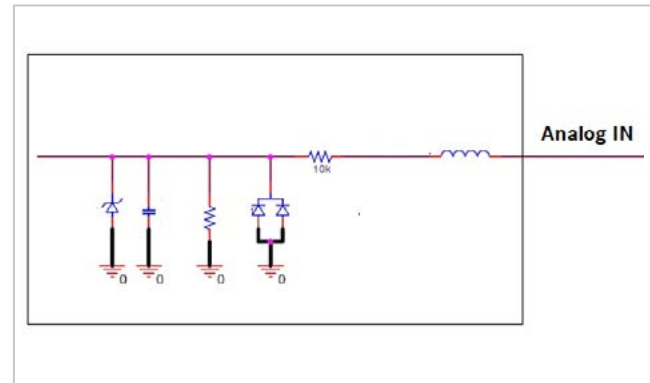
PWM outputs

PWM outputs are used, e.g. to dim a light or to control the speed of single motors. The MBAC can operate up to 14 PWM outputs. The PWM outputs switch to the high side (terminal 30). Of the 14 PWM outputs, 5 can supply a maximum of 5 A and 9 can supply a maximum of 2 A. All the PWM outputs together supply a maximum output of 200 W. The frequency of the PWM is 200 Hz with a tolerance of $\pm 10\%$. The PWM duty cycle can be changed in increments of 0.1 percentage points with a tolerance of $\pm 5\%$.

3.1.3 Analog inputs and outputs

Analog inputs

Analog inputs are used to, e.g. read out the values of a sensor and to recognize voltage thresholds. The MBAC can operate up to 17 analog inputs. These inputs supply an analog voltage between 0.25 V and 10 V with a resolution of 0.1 V and a tolerance of 0.2 V. The input resistance is greater than 5 kOhm.



Analog inputs with pull-up

There are 8 analog inputs with pull-up resistors for the assessment of voltage dividers, e.g. for temperature sensors. There are two sets of inputs each with a of resistance of 1 kOhm and 10 kOhm, and four inputs with a 4.7 kOhm pull-up resistor.

The current is dependent on the value of the internal pull-up and the connected resistor or the value of the temperature sensor (NTC).

Internal pull-up resistors are connected to 5 V.

- 1 kOhm pull-up: Externally connected resistor
 - 1 kOhm: Current = 2.5 mA
- 4.7 kOhm pull-up: Externally connected resistor
 - 4.7 kOhm: Current = 0.53 mA
- 10 kOhm pull-up: Externally connected resistor
 - 10 kOhm: Current = 0.25 mA
- Resolution = 50 mV
- Tolerance = 100 mV

Analog outputs

Analog outputs are used to set voltage values that can, for example, actuate a device that – depending on the voltage applied to its input – executes specific actions. The MBAC can operate up to 5 analog outputs. These outputs supply an analog voltage between 0.25 V and 10 V with a resolution of 0.1 V and a tolerance of 0.2 V. The analog outputs can supply a maximum of 2 mA. The load at the analog outputs must therefore be greater than 5 kOhm. As soon as the read back voltage no longer matches the output voltage, the MBAC switches off the analog output and creates an internal fault entry.

3.1.4 "HSVL+" interface

The MBAC has two "HSVL+ (positive)" interfaces that can be powered through the plug type Rosenberger 59S2UQ-40MT5 and that can be connected to the two displays through a FAKRA coax as a connector. The coding for the first "HSVL+" interface is blue and the second one is green. The interface supports a resolution of 1920 × 720 pixels.

3.1.5 Short-Circuit Proof Capability and Fault Detection

The MBAC can detect the diagnosis states "Open load", "Over temperature"/"Over load" and "Over current". No differentiation is made here between the "Over load" or "Over temperature" diagnosis states. As soon as a fault is detected, the MBAC switches off the affected digital output and creates an internal fault entry. The diagnosis states do not refer here to the rated current, but to the higher – as in the data sheet of the driver used – stated currents.

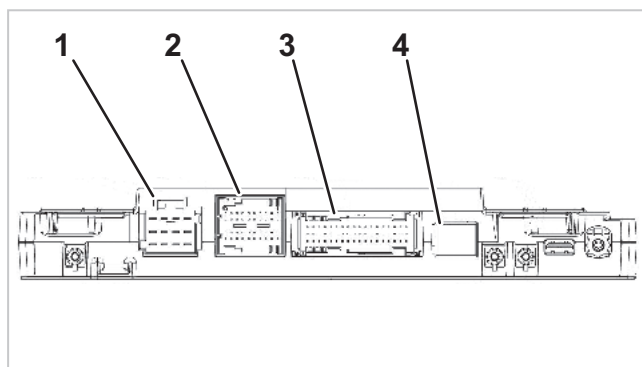
The MBAC switches off a PWM output as soon as a fault ("Open load", "Over load" or "Over current") is detected, and then creates an internal fault entry. The functionality of the PWM outputs is realized through the high-side power switches BTS7004.

3.2 Slots for Body Manufacturer Scopes

3.2.1 MBAC slots on longer side

There are nine slots located on the long side of the MBAC housing. Three of these are FAKRA slots (1×Bluetooth® and 2×HSVL+). Slots for 1×USB and 1×Ethernet (with 4 pins, table 5 (→ page 22)) are also available.

Four additional slots are vehicle plugs for body manufacturers that are presented and described below.



Slots for body manufacturer

- 1 Vehicle plug 6 (with 12 pins) – Table 1
- 2 Vehicle plug 5 (with 40 pins) – Table 2
- 3 Vehicle plug 4 (with 54 pins) – Table 3
- 4 Vehicle plug 3 (with 10 pins) – Table 4

The following tables 1–4 show the defined current and voltage values for each individual pin of the four vehicle plugs.

Table 1 – Body battery, vehicle plug 6 (12 pins)

Pin	Use	Channel	Rated amperage [A]
1	12 V	-	12.5
2	12 V	-	12.5
3	Output, PWM	6	5
4	Output, PWM	7	5
5	Output, PWM	8	5
6	Output, PWM	9	5
7	Output, PWM	10	5
8	Output, PWM	11	2
9	Output, PWM	12	2
10	Output, PWM	13	2
11	Ground	-	12.5
12	Ground	-	12.5

Table 2 – Body vehicle plug 5 (40 pins)

Pin	Use	Channel	Rated amperage [A]
1	Output, PWM	0	2
2	Output, PWM	1	2
3	Digital input *	0	0.01
4	Output, PWM	2	2
5	Digital input *	1	0.01
6	Output, PWM	3	2
7	Digital input *	2	0.01
8	Output, PWM	4	2
9	Digital input *	3	0.01
10	Output, PWM	5	2
11	Analog output	0	0.002
12	Digital input *	4	0.01
13	Digital input *	5	0.01
14	Digital input *	11	0.01
15	Digital input *	12	0.01
16	Digital input *	16	0.01
17	Digital input *	17	0.01
18	Digital input *	8	0.01
19	Digital input *	9	0.01
20	Digital input *	10	0.01
21	Analog output	1	0.002
22	Analog output	2	0.002
23	Digital input *	13	0.01
24	Digital input *	14	0.01
25	Digital input *	15	0.01
26	Digital input	16	0.01

Pin	Use	Channel	Rated amperage [A]
27	Digital input	17	0.01
28	Digital input	18	0.01
29	Digital input	19	0.01
30	Digital input	20	0.01
31	Analog output	3	0.002
32	Analog output	4	0.002
33	Digital input	21	0.01
34	Digital input	22	0.01
35	Digital input	23	0.01
36	Digital input	24	0.01
37	Digital input	25	0.01
38	Digital input	26	0.01
39	Digital input	27	0.01
40	Digital input	28	0.01

* Wake-up capable, digital input

Table 3 – Body vehicle plug 4 (54 pins)

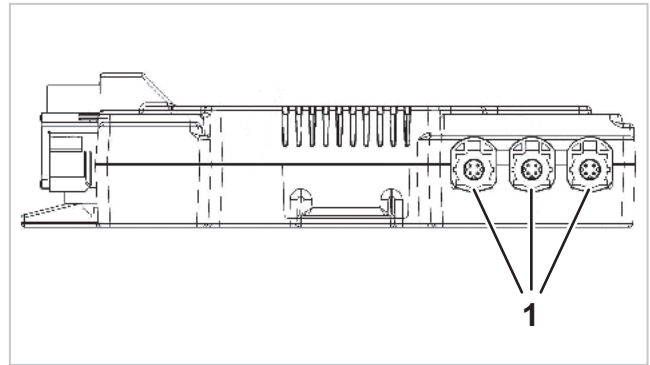
Pin	Use	Channel	Rated amperage [A]
1	MBAC CAN low	–	Additional module
2	MBAC CAN high	–	Additional module
3	Digital output	0	0.2
4	Digital output	1	0.2
5	Digital output	2	0.2
6	Digital output	3	0.2
7	Digital output	4	0.2
8	Digital output	5	0.2
9	Digital output	6	0.2
10	Digital output	7	0.2
11	Digital output	8	0.2
12	Digital output	9	0.2
13	Digital output	10	0.2
14	Analog input	0	0.002
15	Analog input	1	0.002
16	Analog input, pull-up 4.7k	2	0.002
17	Analog input, pull-up 1k	3	0.002
18	Analog input, pull-up 1k	4	0.002
19	CI bus 1	–	–
20	CI bus 2	–	–
21	Digital output	11	0.2
22	Digital output	12	0.2
23	Digital output	13	0.2
24	Digital output	14	0.2

Pin	Use	Channel	Rated amperage [A]
25	Digital output	15	0.2
26	Digital output	16	0.2
27	Digital output	17	0.2
28	Digital output	18	0.2
29	Digital output	19	0.2
30	Digital output	20	0.2
31	Analog input	5	0.002
32	Analog input	6	0.002
33	Analog input	7	0.002
34	Analog input, pull-up 4.7k	8	0.002
35	Analog input, pull-up 4.7k	9	0.002
36	Analog input, pull-up 4.7k	10	0.002
37	Digital output	21	0.2
38	Digital output	22	0.2
39	Digital output	23	0.2
40	Digital output	24	0.2
41	Digital output	25	0.2
42	Digital output	26	0.2
43	Digital output	27	0.2
44	Digital output	28	0.2
45	Coding IO ext. bit 0	–	0.01
46	Coding IO ext. bit 1	–	0.01
47	Ground, analog input	–	–
48	Ground, analog output	–	–
49	Analog input	11	0.002
50	Analog input	12	0.002
51	Analog input	13	0.002
52	Analog input	14	0.002
53	Analog input, pull-up 10k	15	0.002
54	Analog input, pull-up 10k	16	0.002

Table 4 – Body vehicle plug 3 (10 pins)

Pin	Use	Channel	Interface	Rated amperage [A]
1	12 V	-	Vehicle battery	1
2	ABH CAN low	-	MPM	-
3	ABH CAN high	-	MPM	-
4	MBAC CAN low	-	Display	-
5	MBAC CAN high	-	Display	-
6	Digital output	29	MPM wake-up	0.03
7	PWM	14	BT LED push-button switch	0.005
8	Ground	-	BT pushbutton switch	0.01
9	Digital input	29	BT pushbutton switch	0.01
10	Ground	-	Vehicle battery	1

3.2.2 MBAC slots at short side



Slots for body manufacturer (short side)

1 Ethernet slots (3×)

There are 3 more Ethernet slots with 4 pins each on the short side of the housing:

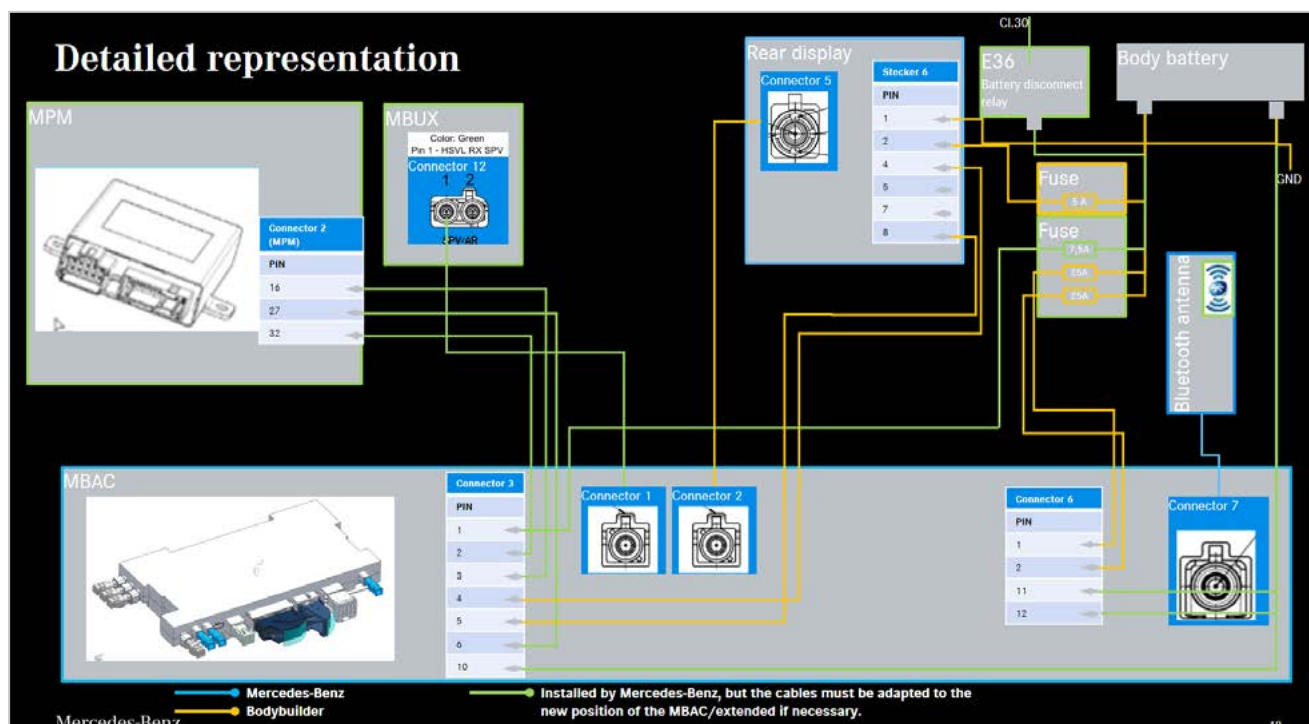
Table 5 – HSD Ethernet plug (4 pins)

Pin	Use	Wire color
1	TX+	Blue
2	RX+	Orange
3	TX-	Green
4	RX-	Brown

The upper part of the MBAC housing has an opening for operation of the DIP switch for the CAN termination (switch is located on the printed circuit board).

Located at the left of the opening is the switch position information in a recessed field ($t = 0.5 \text{ mm}$) on the upper part of the MBAC housing.

3.2.3 Block diagram of slots



Plug 2 (PSM/MPM)

Pin	Use	Assignment
16	Body manufacturer CAN (CAN-ABH)	CAN ABH_H
27	Multifunction input 03	High side (IN_3)
32	Body manufacturer CAN (CAN-ABH)	CAN ABH_L

Plug 6 (additional display)

Pin	Use	To interface
1	Ground	Terminal 31
2	12 V	Terminal 30
4	CAN low	MBAC, plug 3 pin 4
5	UART_RX	n.a.
7	UART_TX	n.a.
8	CAN high	MBAC, plug 3 pin 5

Plug 3 (MBAC)

Pin	Use	To interface
1	12 V	Terminal 30
2	ABH CAN low	MPM, plug 2 pin 32
3	ABH CAN high	MPM, plug 2 pin 16
4	MBAC CAN low	Additional display, Plug 6 pin 4
5	MBAC CAN high	Additional display, Plug 6 pin 8
6	PSM (MPM) wake-up	PSM (MPM), Plug 2 pin 27
10	Ground	Plug 3

Plug 6 (MBAC)

Pin	Use	Interface
1	12 V	Terminal 30
2	12 V	Terminal 30
11	Ground	Terminal 31
12	Ground	Terminal 31

4.1 Peripheral Devices

The following standard interfaces are supported:

- MBAC-CAN
- ABH CAN
- CI bus
- HSVL+
- Digital inputs
- Digital outputs
- Analog inputs
- Analog outputs
- USB
- Power supply
- Contacting/plugs

4.1.1 CAN

The MBAC has two CAN interfaces: one MBAC-CAN and one ABH-CAN interface with a transfer rate each of 250 kBit/s. The CAN interface corresponds to CAN-HS (CAN High Speed) and it guarantees a maximum physical transfer rate of 1 MBit/s. The physical layer of the CAN interfaces is implemented as per A2130006299. The termination of the MBAC-CAN can be changed using the DIP switch to 60 ohms or 120 ohms. The DIP switch for configuring the previously-mentioned termination is located in an opening of the upper part of the MBAC housing.

4.1.2 LIN/CI

The MBAC has two LIN/CI interfaces that guarantee a physical transfer rate of 20 kBit/s. The LIN interfaces are provided as per A1660006899 and they are designed to be a master. The standard for the LIN transceiver used is LIN 2.2 A.

4.1.3 USB

The MBAC is updated via the USB interface. This interface is provided exclusively for these software updates and must not be used by the body manufacturer.

4.1.4 Ethernet

The MBAC contains 4 Ethernet interfaces for external connections that all support auto MDI/MDIX features.

The standard for the Ethernet interfaces corresponds to Fast Ethernet (100BASE-TX) and the coding for the 4 HSD plugs is black. The assignment of the Ethernet HSD plug can be found in Chapter 3.2.2 MBAC slots at short side (→ page 22).

4.1.5 Ethernet activation and deactivation

The changes of the MBAC (Basic/Stage 2 equipment variant) into its possible operating modes and the effects on the Ethernet (activation or deactivation) are described below.

- "External voltage" operating mode
 - Every deactivated Ethernet interface on the MBAC is activated.
- "Body's own voltage" operating mode
 - Every deactivated Ethernet interface on the MBAC is activated.
- "ECO body's own voltage" operating mode
 - Every active Ethernet interface is deactivated.
- "ECO external voltage" operating mode
 - Every active Ethernet interface is deactivated.
- "Undervoltage" operating mode
 - Every active Ethernet interface is deactivated.
The Bluetooth® interface must be activated in the "Undervoltage" operating mode so that people using the MBAC App via smartphone or tablet are notified about any undervoltage state that exists.
- "Off" operating mode
 - Every active Ethernet interface is deactivated.
 - The MBAC searches for appicals-compatible devices and detects any appicals-compatible device connected through the Ethernet interface. If this device logs out over the Ethernet interface, it is then deleted from the memory by the MBAC. The MBAC also routes the appicals messages between the MBAC-CAN interface and each Ethernet interface as well as between the Bluetooth® interface and each Ethernet interface.
To ensure that the MBAC is disconnected from the Ethernet network in the vehicle, VLAN tags are provided in combination with a PDU router.

4.2 Supported peripherals

To be able to be connected to the MBAC, the components from other manufacturers are expected to have corresponding physical interfaces. As an example, for the climate control, the Hydronic, Airtronic and CI bus heating and the CI bus air conditioning system components could be connected to the MBAC.

CI bus

CI bus systems are used to control CI-compatible devices and functions in the camper van's on-board electrical system. It is not just the lighting or charging process, but also many other components installed in the on-board system such as the air conditioning system or heating that can be centrally, easily and reliably operated using the CI bus.

 For information on the use of CI devices, please get in touch with the contact persons from the technical advice service for the MBAC (→ page 9).

The available LIN/CI bus signals result from the configuration options of the models (see Chapter 4.2.6 Description of models (→ page 28)). Not all LIN/CI bus signals are made available by the configuration.

4.2.1 Negative Temperature Coefficient thermistor (NTC)

An NTC can be connected to an analog input of the MBAC for measurement of the temperature in the vehicle. The actual measured temperature may be higher or lower.

4.2.2 Light control

High-side lamps can be used as dimmable lights or as color-temperature or color-controllable lights. "Tunable-Light", "DimmableLight" or "ColorLight" models can be used here for actuation.

4.2.3 Climate control

The climate control can be represented here by the "ClimateControl" model.

Hydronic

Hydronic is an integrated device used for regulating the temperature in three different areas of the vehicle and for generating hot water. It also controls the boiler temperature and opens or closes the valves of the heating areas.

Airtronic

Airtronic is an integrated device used to regulate the temperature in the vehicle.

CI bus heater and air conditioning system

A CI bus heater and CI bus air conditioning system can be connected for the climate control.

GPIO roof ventilator

A GPIO roof ventilator can be actuated by the MBAC to regulate the interior climate. The control system for a GPIO roof ventilator, and therefore also the GPIO roof ventilator itself, do not have independent characteristics. The "WritableValue" can be used for actuation.

4.2.4 Comfort components

Pop-up and/or sliding sunroof

A pop-up roof can be opened upwards for comfort purposes to allow a standing height in the interior. The sliding sunroof is a roof system that can be opened. As this involves the actuation of motor components, the "MechanicalControl" model is used by preference here.

GPIO awning

A GPIO awning can be actuated by relay for a higher current load. The "MechanicalControl" model can also be used here for actuation.

SAT system

A SAT system can be connected to the MBAC to provide TV reception via satellite. The MBAC is only responsible here for control of the SAT system and it does not provide a TV signal. The SAT system automatically aims at the satellites. A suitable model here for actuation is the "WritableValue" model.

CI refrigerator

A CI-compatible refrigerator connected to the MBAC can be actuated through the MBAC.

It is possible to set the cooling level or the switching on and off of the device, as well as other functions that are provided via the CI bus.

GPIO refrigerator box

The temperature of a GPIO refrigerator box is regulated in a required range.

TV drawer

A TV drawer can be connected to the MBAC through a relay for retracting and extending. This motor control can be implemented using the "MechanicalControl" model.

GPIO lifting bed

A GPIO lifting bed can be connected to the MBAC for comfort purposes. The "MechanicalControl" model can also be used here for actuation.

Slide-out (extendable body window)

A slide-out can be connected to the MBAC for the status request. The "ReadOnlyValue" model can be used to process the values.

4.2.5 Supply components

5-bar probe

A 5-bar probe can be read in by the MBAC over analog inputs, to record the current fill levels of the gray water and fresh water tanks.

2-bar probes are also supported, as well as capacitive measuring probes.

GPIO water drain valve

The MBAC can open and close a connected GPIO water drain valve (MechanicalControl model).

GPIO water pump

The MBAC can activate a GPIO water pump for pressure buildup in the coolant lines.

GPIO gas shutoff valve

A GPIO gas shutoff valve can be connected to the MBAC to regulate the supply of gas. The GPIO gas shutoff valve can be actuated by the MBAC through either:

- a digital output with relay in the wiring harness or
- a PWM output.

GPIO gas fill level sensor

A GPIO gas fill level sensor can be mounted to the outside of a gas bottle and connected to the MBAC to monitor the fill level.

4.2.6 Description of models

The models mentioned in the preceding chapters will be described in greater detail below.

ClimateControl

The ClimateControl model is used to control air conditioning systems and heating systems.

Depending on the components involved, the following functions are available:

- Switch air-conditioning unit on/off
- Set target temperature,
 - within a defined interval [T_min, T_max]
 - through stages
- Set fan speed,
 - continuously variable
 - through stages
- With heaters, e.g. select the energy source and the heating output to be generated
- Set function mode for CI components

If the Airtronic temperature sensors malfunction, substitute values can be defined.

Control can take place through CI and GPIO.

For GPIO, the following devices are available:

- Airtronic
- Hydronic

The Airtronic control system uses

- 1×DO for activation,
- 1×AO for target voltage for target temperature,
- 1×AO for reference voltage for specified temperature,
- 1×AI for fault diagnosis,
- 1×AI for temperature measurement (either with external NTC pull-up or internal NTC pull-up)

and it is implemented as in the Airtronic specification.

The Hydronic control system uses

- 1×DO for boiler activation,
- 1×DO for activation of Plugtronic,
- 1×PWM for fan speed,
- 2×PWM for energy supply for fan in bathroom and living area,
- 1×PWM for energy supply of valves,
- 3×AI for temperature measurement (either with external NTC pull-up or internal NTC pull-up) of the various heating zones,
- 6×DO for valve control

and it is implemented as in the Hydronic specification.

For CI, the following devices are available:

- Dometic Freshjet A/C
- Truma Air Conditioning
- Alde Heater 3020
- TrumaCombiSystem

The control system is implemented as per the CI specifications.

As a function extension for air conditioner units, a timer control or events for additional models enable the following to be configured and queried during the runtime:

- Control system active
- Target temperature
- Energy source
- Output
- Fan speed

The model also provides the following properties:

- Current temperature
- Air conditioning system active
- Heater active

ColorLight

The ColorLight model enables control of the RGB lights with three channels.

Depending on the components involved, the following functions are available:

- Switch lights on/off
- Switch lights on/off using pushbutton switch
- Dim lights by pressing button longer
- Regulate color component through PWM duty cycle for each channel
- Dim through ramp (using specified time)
- Group the lights

Automatic control depending on the given conditions (e.g. signal from motion detector present)

Control is performed through GPIOs and uses 3×PWM outputs (1× for each channel).

As a function extension for RGB lights, for additional models, a timer control or events enable the following to be configured and queried during the runtime:

- Control system active
- Channel value red
- Channel value green
- Channel value blue

Lighting scenarios and model events can also be defined here that serve to manipulate the lights in general and light groupings.

DimmableLight

The DimmableLight model can be used to control dimmable lights.

Depending on the components involved, the following functions are available:

- Switch lights on/off
- Switch lights on/off using pushbutton switch
- Dim lights by pressing button longer
- Dim through ramp (using specified time)
- Group the lights

Automatic control depending on the given conditions (e.g. signal from motion detector present).

Control is performed through GPIOs and uses 1×PWM output.

Lighting scenarios and model events can also be defined here that serve to manipulate the lights in general and light groupings.

Key

The Key model enables the pushbutton switches to be read in.

The following detection options are available:

- Pushbutton switch pressed
- Pushbutton switch pressed for long period
- Pushbutton switch blocked
- Pushbutton switch released
- Pushbutton switch was blocked but not anymore
- Pushbutton switch clicked

The read-in takes place via 1×DI.

As an additional function, button presses can be used as events for model changes (e.g. for the dimming of lights or the extension and retraction of awnings).

MechanicalControl

The MechanicalControl model can be used to control

- (component) motors with relays and "active direction" control, i.e. one control input for activation and one control output for the motor direction,
- (component) motors with "In-Out" control, i.e. one control input for retracting and one control output for extending a component.

The following functions are available:

- Provision of a continuous control signal until an end position is reached (e.g. by continuously pressing the button or through timer control)
- One-time signal to initiate motor movement
- Timeouts, after which the safe or functional end position is presumed to have been reached and the motor movement stops
- Reading in of the status of limit switches in the safe and/or functional state
- Automatic retracting/extending or suppression of function as soon as defined conditions are fulfilled (e.g. vehicle in motion).

Control is performed through GPIOs and uses:

- 2×DO for motor movement
- Optional 1×DO for signal to stop motor movement
- Optional 2×DI for limit switches

The following commands can be executed as a function extension for the MechanicalControl model during the runtime:

- Switch into safe status
- Switch into functional status
- Stop

The MechanicalControl model provides the following properties as events for the behavior control:

- Status
- Control mode

Some application examples for MechanicalControl include:

- Retracting, extending, tilting the sliding sunroof
- Retracting, extending the step

Additional devices with a similar functional scope include:

- TV drawer
- Lifting bed
- Awning
- Pop-up roof
- Valve control

OnOffControl

The OnOffControl model enables refrigerator boxes to be controlled using On/Off logic depending on the given temperature values and using components with On/Off logic.

The following functions are available:

- Configuration of a minimum switch-off time before the component can be activated again
- Hysteresis of sensor value
- Set target value (continuous, in stages)
- Activate/deactivate component

If a GPIO refrigerator box is involved, the following is used for the control system:

- 1×DO for activation (compressor)
- 1×DO for NTC pull-up activation
- 1×AI with look-up for current temperature

If a CI component is involved, the following can be changed during the runtime (provided that the CI component supports the function):

- Select energy mode
- Query current energy mode
- Query availability of automatic mode
- Query temperature status
- Query active user interaction

For GPIO refrigerator boxes, monitoring is carried out for the following faults during the runtime:

- Temperature sensor defective or not connected
- Temperature sensor above permitted limit
- Temperature sensor below permitted limit

As a function extension for the OnOffControl model, for additional models, a timer control or events enable the following to be configured and queried during the runtime:

- Control system active
- Target value

The model also provides the following properties for the behavior control:

- Currently read-in value
- Output active

ReadOnlyValue

The ReadOnlyValue model enables contacts, fill levels, (relative) sensor values and logical states to be read in.

The following functions are available:

- Data about minimum change before this is communicated
- Conversions of read-in values
- Read in of fault states for CI devices

Reading-in can be performed through GPIO, CI and CAN.

For GPIO, the following connections can be used for reading in:

- 1×AI
- 1×AI with conversion formula
- 1×DI
- 1×DO and 4×AI for conventional fill level measurement
- 1×AI for capacitive fill level measurement
- 4×DO and 1×AI for inverted fill level measurement

 For information on the use of CI devices, please get in touch with the contact persons from the technical advice service for the MBAC (→ page 9).

The following information can be read in for the CAN:

- Doors open/closed
- Doors locked/unlocked
- Tank cap open
- Ignition status
- Engine running
- Vehicle speed
- Temperature unit
- Speed unit
- Language setting
- Vehicle battery voltage and current

For cyclical measurements (e.g. fill levels) the model provides the following options:

- Activate cyclical measurement
- Time interval for idle state (no measurement)

One model application example (with behavioral control) is

- Motion detector detects, motion and lights are switched on for defined time interval.

An analog fuel level device can be configured. To do so, the read-in voltage value U_{max} , which is equivalent to a 100 % fill level, must be known. A table with x control points in the interval $[0 \text{ V}, U_{\text{max}}]$ and the corresponding fill levels is then filed in the configuration (equivalent to the variant 1x AI with conversion).

TuneableLight

The TuneableLight model enables the control of color temperature-controllable lights to be implemented, which is made possible by one channel each for warm white/cool white.

Depending on the components involved, the following functions are available:

- Switch lights on/off
- Switch lights on/off using pushbutton switch
- Dim lights by pressing button longer
- Set color temperature (warm white/cool white)
- Dim through ramp (using specified time)
- Group the lights
- Automatic control depending on the given conditions (e.g. signal from motion detector present)

Control is performed through GPIOs and it uses

- 1x PWM output for warm white
- 1x PWM output for cool white

As a function extension for the TuneableLight model, a timer control or events for additional models enable the following to be configured and queried during the runtime:

- Control system active
- Brightness target value
- Color temperature target value

Lighting scenarios and model events can also be defined here that serve to manipulate the lights in general and light groupings.

WritableValue

The WritableValue model enables control of the following:

- On/Off state for single devices
- Switching of light groups

Reading-in can be performed through GPIO, CI and CAN.

For GPIO, the following connections can be used for setting:

- 1×AO
- 1×AO with conversion formula
- 1×DO

The following information can be set for the CI:

- Alde Heater 3020
 - Hot water setting
- Teleco SAT Easy/Active
 - Satellite search
 - Retract satellite dish

If the WritableValue model is to be used to control light groups, the interfaces are assumed to comply with the descriptions of the individual models for dimmable/RGB/color temperature-controllable lights.

As a function extension for the WritableValue model, for additional models, a timer control or events enable the target value to be configured and queried during the runtime.

One application example for using the model and querying the target value is the condition whereby a motion detector must be actively switched to trigger an alarm.

Battery

The Battery model enables the sensor values of an intelligent battery management system (IBS) to be read in and the fault states on CI devices to be read in.

Depending on the components involved, the following functions are available:

- Set battery type (starter/gel/AGM)
- Nominal battery capacity

IBS values can be provided for the following components for the CI:

- Battery voltage
- Battery current
 - Remaining time in minutes
 - State of Charge (SoC)
 - State of Health (SoH)

The following faults can be managed in the central fault memory:

- Hella IBS
 - Sulfation
 - Battery defective
 - Fault in CI communication

The remaining time is calculated internally using the Peukert formula based on the battery current and the battery type.

The Battery model does not provide any behavioral control methods.

AlarmFunctions

The AlarmFunctions model enables the functionality and management of alarms.

The following functions are available:

- Data on sensor values to be read in
- KO gas
- Smoke
- LPG gas
- Data on triggered alarms
- Horn
- Exterior lighting
- Headlamps

The devices must be configured with suitable models to ensure the model's functionality is given.

UserMessage

The UserMessage model enables the trigger an event message (eventMessage) functionality and the trigger a state message (stateMessage) functionality.

The following functions are available:

- Determination of ID of triggering message (eventMessage or stateMessage)
- Determination of a time interval before a message is triggered again
- Determination of conditions that must be fulfilled to ensure that the message is triggered.
- Management of a maximum of 16 event messages (latest event messages delete the oldest ones when the limit is reached)
- State messages cannot be deleted.

Messages should inform the user about the status of the vehicle.

The conditions refer to model values for other models (e.g. ReadOnlyValue).

5.1 General information

The MBAC can also be controlled using the MBAC App. The MBAC app can provide the same functional scope as the head unit.

The MBAC app is connected to the MBAC via Bluetooth®, for which – depending on the surrounding conditions and the positioning of the Bluetooth® antenna – the working range is 15 m. This gives the customer added comfort as the functions installed in the MBAC can be operated when they are in the Bluetooth® range.

An Internet connection is not required when using the MBAC app.

The MBAC app is available to download from the App Store.

The MBAC app is free and is regularly updated.

5.2 Setup

When installing the MBAC App for the first time, the customer is guided through a setup process. In this process, the customer is notified about the terms of use for the MBAC App. The basic settings are also made here. The use of Bluetooth® must be permitted here for the MBAC app, since it is a basic requirement for using the app.

When using the MBAC App for the first time together with the MBAC, Bluetooth pairing must be configured first. To do so, the MBAC in the vehicle must be set to visible (MBAC Menu/Settings/Bluetooth®). After this, the MBAC is visible in the Bluetooth® menu of the MBAC App. Pairing is started by selecting the MBAC in the Bluetooth® menu of the app.

The customer must then run through a two-factor MBAC authentication process:

- Firstly, the last 6 digits of the vehicle identification number (FIN) for the vehicle in which the MBAC is installed are required.
- Secondly, the last 8 characters (digits or letters) of the vehicle identification number (FIN) for the vehicle in which the MBAC is installed are required.

Once successfully paired, the MBAC App downloads the configuration file from the MBAC. The customer does not have to do anything here. However, during the download, an information view is shown in the MBAC App until the process has completed successfully. The customer is then automatically forwarded to the MBAC App and can then use the functions.

▶ Before shipment, the displayed functions should be compared with the ones in the configuration. This check can also be performed using the view on the head unit.

6.1 Commissioning

Process

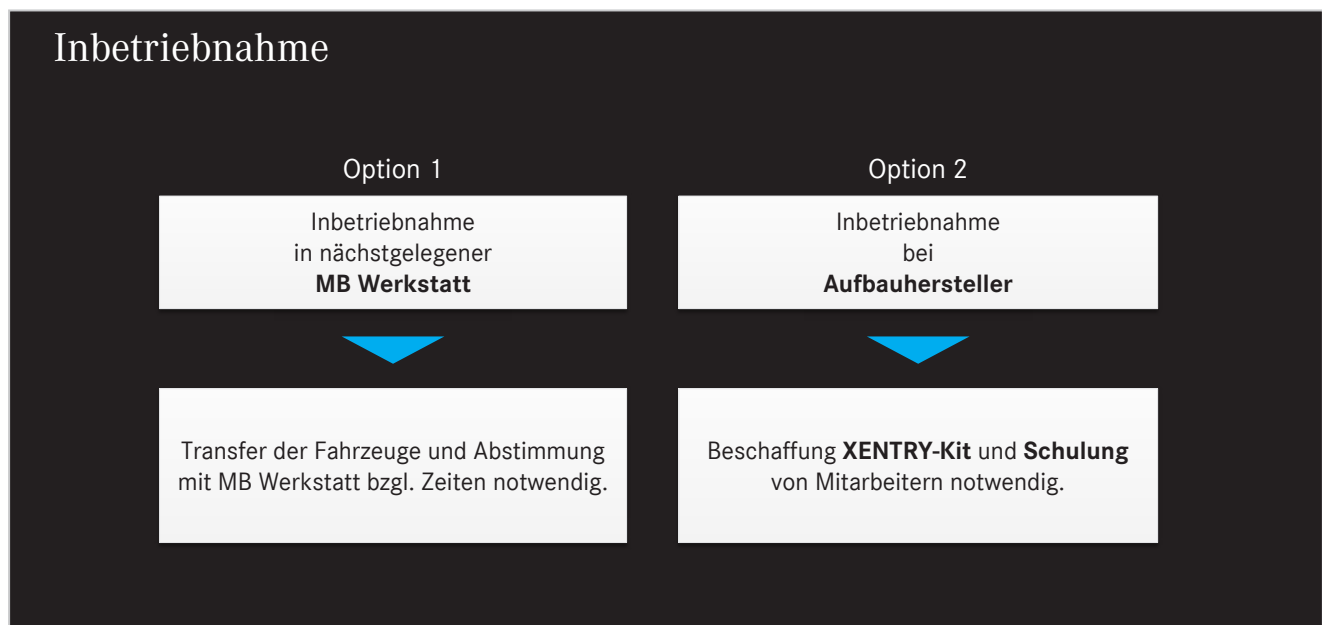
The commissioning process contains the following steps:

- Programming (software update)
- SCN coding
- Transfer configuration

Location

Commissioning can be performed either in a Mercedes-Benz workshop (Option 1) or directly at the body manufacturer's (Option 2).

For conditions or advantages/disadvantages, see following diagram.



6.1.1 Coding and configuration

A Config file enables the MBAC to execute functions and to control components. This Config file must be created to match the vehicle or body.

The Config file must then be transferred to the MBAC control module with the aid of XENTRY Kit. For information on the XENTRY Kit, see Chapter 6.2.3 XENTRY Kit, Diagnosis (→ page 39).

Mercedes-Benz/body manufacturer responsibilities

The Config file is provided by Mercedes-Benz in cooperation with the body manufacturer in accordance with its specifications in terms of menu structure, icons and texts.

Depending on the commissioning location, the Config file is transferred using XENTRY Kit at Mercedes-Benz or by the body manufacturer.

End-of-line test

The functionality of the MBAC config file is checked in collaboration with the body manufacturer and Mercedes-Benz.

The body manufacturer is obliged to test and document the functionality of the MBAC in accordance with the config definition in combination with the installed terminals and functions during the end-of-line test.

Responsibilities of Mercedes-Benz

- Vehicle
- MBAC (control unit)
- Config file for configuring the MBAC
- XENTRY Kit

Responsibilities of body manufacturer

- MBAC peripherals (wiring harness)
- Terminals and functions

6.2 Service

6.2.1 General information

The end-installation location selected freely by the body manufacturer and the free configuration mean that the MBAC Aftersales responsibility lies fully with the body manufacturer.

6.2.2 Tests Conducted by the Body Manufacturer

The body manufacturer is at all times the first point of contact.

Steps to be conducted by the body manufacturer:

- ▶ Test of terminals
- ▶ Test of components and wiring harness

MBAC

- ▶ Test of power supply (wiring harness, fuses, contacting)
- ▶ Conductance of a diagnosis at the body manufacturer's or in a Mercedes-Benz workshop
- ▶ If a defect is detected, order a new component part through the Mercedes-Benz workshop/dealer and install at the body manufacturer.

Additional display

- ▶ If a defect is detected or if visible damage is found, order new component part through the Mercedes-Benz replacement part system.
- ▶ Complaints processed through Mercedes-Benz dealers

6.2.3 XENTRY Kit, Diagnosis

The XENTRY Kit diagnosis tool can be used to parameterize and update the MBAC and to perform diagnosis. This process does not differ from other control units installed in the vehicle.

It is assumed that you are familiar with the use of XENTRY Kit.

- ❗ Further information on XENTRY Kit and Diagnosis can be found in the vehicle-specific body/equipment mounting directive and at:

<https://b2bconnect.daimler.com>

- ❗ Information on possible training offers can be found online at:

<https://mobile.gt.mercedes-benz.com>

6.2.4 Removal/Installation

The MBAC Removal and Installation Instructions must be compiled by the body manufacturer and made available as and when requested by the Mercedes-Benz organization.

Tools

Special tools should not be used for removal and installation of the control unit.

Replacement parts

Replacement parts can be procured through the Mercedes-Benz replacement parts system (Over the Counter).

6.2.5 Warranty

The genuine Mercedes-Benz scope of supply is subject in full to the warranty terms of Mercedes-Benz AG. Warranty claims for a defective MBAC are to be submitted to an authorized Mercedes-Benz service partner.

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List of abbreviations

Abbreviation	Explanation
ABH	Body manufacturer
AeJ	Modification year (YoM)
AI, AO	Analog Input or Output
ARL	Body/Equipment Mounting Directive(s)
BR	Model series
CAN	Serial bus system / Controller Area Network
CI bus	Special bus system for camper vans for the camping industry (CI)
DI, DO	Digital Input or Output
DIP switch	Switch on printed circuit boards with two parallel connection rows (dual in-line package)
EG	European Community (EC)
FAKRA	DIN Standards Committee for Motor Vehicles, special connector for on-board electronics
FIN	Vehicle identification number (VIN)
GND	Ground connection
GPIO	Programmable inputs and outputs / General Purpose Input Output
HSD	High-speed data transfer / High Speed Data
HSV L+	Coaxial High Speed Video Link (for video signal transfer) for automotive applications
ISO	International Organization for Standardization
KO gas	Narcotic gases, often used by criminals to daze campers
LIN	Local Interconnect Network
LPG	Mixture of propane and butane / Liquefied Petroleum Gas
MBAC	Mercedes-Benz Advanced Control, control unit and interface module for camper vans
MBUX	Mercedes-Benz User Experience, head unit multimedia system
MPM	Multi-Purpose Module, Mercedes-Benz-internal designation for PSM
NTC	Temperature sensor / Negative Temperature Coefficient thermistor
PSM	Parameterizable Special Module / Multi-Purpose Module (MPM)
PWM	Pulse Width Modulation, square wave signal with constant cycle duration
RGB	Color space from the basic colors Red, Green and Blue
SCN	Parameterized number for coding a control unit / Software Calibration Number
T	Temperature
USB	Data transfer system between a computer and external devices / Universal Serial Bus
VIN	Vehicle Identification Number

